

AL BURAIMI, Oman

WMO: 412440

Lat: 24.2337N

Lon: 55.9155E

Elev: 1220

StdP: 14.06

Time Zone: 4.00 (E04)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.0	53.5	21.2	16.6	82.8	24.9	19.8	86.2	21.5	72.9	18.6	70.8	2.5	130	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.4	112.8	69.4	111.3	69.6	109.7	69.8	80.7	92.0	79.3	93.2	77.9	94.2	9.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	152.3	87.5	75.7	140.7	88.6	73.1	128.8	88.8	45.6	91.9	44.0	93.1	42.5	94.1	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	17.1	15.1	DB	46.2	115.6	3.5	1.1	43.7	116.3	41.7	117.0	39.7	117.6	37.2	118.4	(4)
(5)			WB	40.3	83.0	2.9	1.4	38.3	84.1	36.6	84.9	35.0	85.7	32.9	86.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.7	65.1	68.6	75.2	84.0	92.2	96.9	98.8	98.7	93.7	86.1	75.8	68.2		
	DBStd	12.83	4.33	5.62	5.98	5.67	4.26	3.67	3.65	2.79	3.19	3.32	4.13	4.11		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	96	52	23	3	0	0	0	0	0	0	0	0	18		
	CDD50	12297	469	522	780	1020	1308	1406	1513	1510	1310	1120	774	565		
	CDD65	6917	56	125	317	570	843	956	1048	1045	860	655	324	118		
	CDH74	105965	329	1074	3493	7799	13559	16280	18195	18197	14028	9174	3116	721		
	CDH80	71685	22	282	1563	4605	9373	11997	13740	13734	9751	5359	1168	91		
(14) Wind	WSAvg	7.0	6.2	7.0	7.4	7.3	7.1	7.7	7.9	7.8	7.4	6.7	6.0	5.9		
(15) Precipitation	PrecAvg	3.5	0.4	0.7	0.8	0.4	0.0	0.1	0.3	0.2	0.1	0.0	0.1	0.3		
	PrecMax	10.4	2.1	3.0	3.4	2.4	0.2	1.1	4.0	0.6	1.4	0.4	0.7	2.0		
	PrecMin	0.7	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0		
	PrecStd	2.3	0.6	0.7	1.1	0.6	0.0	0.2	0.7	0.1	0.2	0.1	0.2	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.7	89.8	97.2	104.7	112.3	114.9	114.5	113.7	110.4	103.5	93.6	85.1		
		MCWB	60.3	61.5	61.5	65.3	66.7	69.5	70.1	69.9	69.0	66.2	64.3	61.9		
	2%	DB	79.8	86.4	94.2	102.0	108.8	112.5	112.7	111.7	107.9	101.0	91.0	82.5		
		MCWB	60.1	60.8	61.3	64.9	66.8	68.5	70.2	70.3	68.6	65.9	64.0	61.7		
	5%	DB	77.7	83.6	91.5	99.7	106.7	110.7	110.9	110.0	106.0	99.0	89.0	80.4		
		MCWB	59.3	60.2	61.2	64.2	66.9	68.5	70.6	70.6	68.7	65.7	63.8	61.6		
	10%	DB	75.5	80.6	88.6	97.2	104.8	108.8	109.1	108.4	104.2	96.9	86.8	78.3		
		MCWB	58.3	59.7	60.6	63.6	66.7	68.7	71.0	70.9	69.0	65.3	63.7	61.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	67.5	69.9	71.7	75.6	80.8	82.5	82.4	80.7	76.9	71.9	67.8		
		MCDB	71.2	79.9	81.7	89.5	90.6	91.2	90.4	92.2	88.8	84.8	79.2	74.8		
	2%	WB	63.8	65.6	67.7	69.4	73.5	78.3	80.8	80.5	78.8	74.7	70.1	66.1		
		MCDB	72.3	76.7	80.8	88.4	92.8	92.9	92.0	93.5	88.6	84.3	79.0	75.1		
	5%	WB	62.3	64.2	65.9	67.9	71.5	76.5	79.4	78.8	77.1	73.4	68.7	64.8		
		MCDB	72.7	75.6	80.8	88.1	94.4	93.5	93.8	95.7	90.0	84.0	79.0	73.9		
	10%	WB	60.9	62.6	64.2	66.5	69.8	75.0	78.0	77.0	75.3	71.9	67.1	63.6		
		MCDB	72.0	75.0	78.9	87.2	94.9	94.4	95.5	97.2	91.4	85.2	79.0	73.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.8	22.5	22.7	24.3	24.1	20.9	20.4	21.4	21.6	19.8	18.5		
		MCDBR	20.9	23.3	25.1	25.1	26.3	25.3	22.2	21.7	22.5	23.3	21.7	20.7		
		MCWBR	8.2	9.0	8.9	8.5	8.7	8.4	8.1	8.1	9.3	9.9	8.8	8.3		
		MCDBR	17.1	18.0	18.6	20.5	23.6	23.9	21.5	21.2	22.3	22.5	20.6	17.6		
	5% WB	MCWBR	8.1	8.5	7.9	8.6	9.3	9.0	8.6	9.1	10.5	10.0	8.6	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.413	0.466	0.529	0.569	0.571	0.635	0.720	0.686	0.583	0.480	0.427	0.408		
	taud		2.112	1.948	1.786	1.728	1.721	1.620	1.519	1.572	1.730	1.940	2.096	2.125		
	Ebn at Noon		264	257	247	240	239	222	204	211	230	249	257	259		
	Edh at Noon		45	56	70	75	76	83	92	87	73	56	45	43		
(44) All-Sky Solar Radiation	RadAvg		1389	1632	1910	2147	2381	2402	2223	2165	2036	1804	1481	1318		
	RadStd		76	101	116	131	60	67	81	55	39	37	32	66		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)		+0.85	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	+270	+268

Nomenclature: See separate page